Optimized Multistage Decimation Based on Optimal Factorization of Decimation Ratio

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Introduction

Changing the sampling rate is a fundamental task in multirate digital signal processing. For prime, non-integer, or poorly factorizable decimation ratios, conventional FIR-only systems may become computationally inefficient. This work proposes a generalized multistage decimation structure in which:

- the first stage performs non-integer decimation using a Transposed Modified Farrow Structure (TMFS),
- the remaining stages perform integer decimation using FIR filters,
- the overall factorization is chosen to minimize the computation rate, measured in multiplications per second.

Generalized multistage decimation

The overall decimation ratio is expressed as $R = \beta \prod_{k=1}^{K} N_k$, where:

- β is a non-integer decimation factor implemented by TMFS,
- N_k are integer decimation factors implemented by FIR stages.

The equivalent transfer function is $H(z) = H_\beta(z) \prod_{k=1}^{K} H_k\left(z^{\widehat{N}_k}\right)$, $\quad \widehat{N}_k = \prod_{i=1}^{k-1} N_i$.

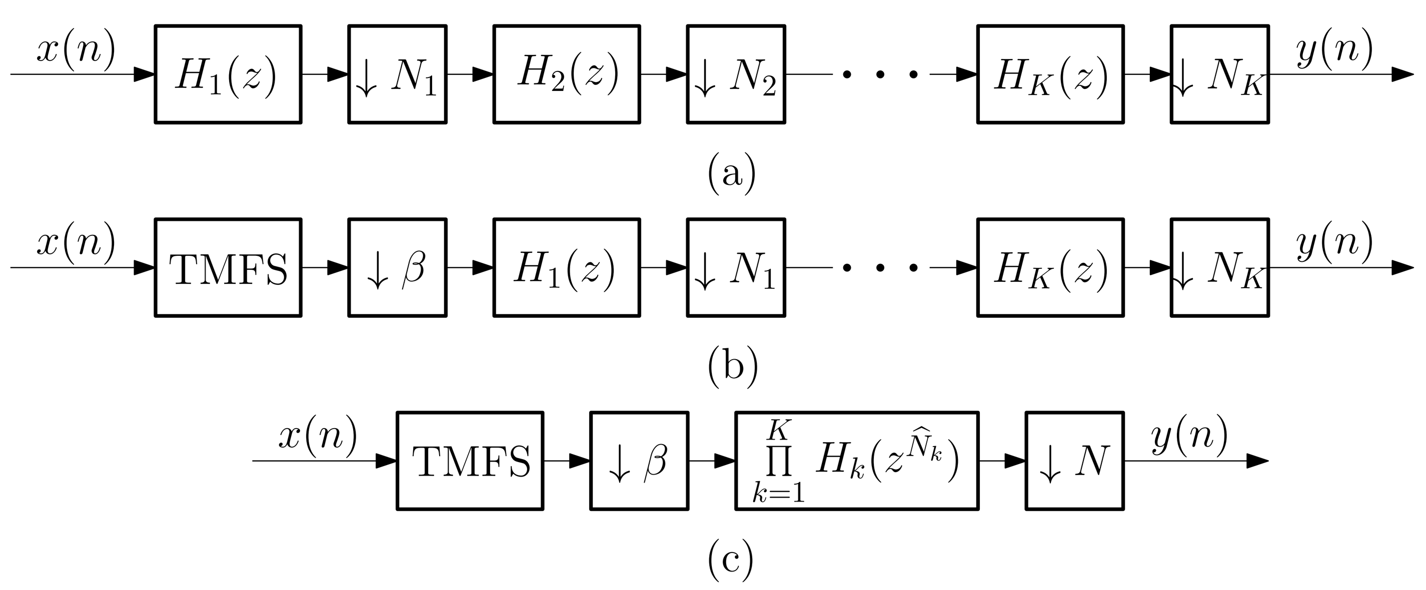

Figure 1. (a) Conventional multistage FIR decimator, (b) Cascaded TMFS and conventional multistage FIR decimator, and (c) Equivalent representation of the structure (b).

Passband and stopband constraints

For the last FIR stage ($k = K$): $\quad \Omega_p^{(K)} = [0, f_p/N_K], \quad \Omega_s^{(K)} = [1/N_K, 1]$.

For intermediate FIR stages ($k = 1, \ldots, K-1$):

$$\Omega_p^{(k)} = [0, f_p \widehat{N}_k/N], \qquad \Omega_s^{(k)} = \bigcup_{l=1}^{\lfloor N_k/2 \rfloor} \left[\frac{2l - \widehat{N}_{k+1}/N}{N_k}, \min\left(\frac{2l + \widehat{N}_{k+1}/N}{N_k}, 1\right)\right].$$

For the TMFS stage:

$$\Omega_p^{(\beta)} = [0, f_p/(2N)], \qquad \Omega_s^{(\beta)} = \bigcup_{l=1}^{\lfloor \beta/2 \rfloor} \left[l - \frac{1}{2N}, \min\left(l + \frac{1}{2N}, \beta/2\right)\right].$$

Transition bandwidths therefore are:

$$\Delta f_\beta = \frac{2N - (1+f_p)}{2N}, \qquad \Delta f_k = \frac{2 - \frac{\widehat{N}_{k+1}}{N}(1+f_p)}{N_k}, \qquad \Delta f_K = \frac{1-f_p}{N_K}.$$

Computation rate analysis

The ripple specifications for each subfilter are $\delta_p/(K+1),\ \delta_s$.
The total computation rate is $C_T = C_\beta + C_{\mathrm{FIR}}$.
For the TMFS stage

$$C_\beta = F_{\mathrm{in}} M_\beta + F_{\mathrm{in}} \frac{N}{R}\frac{L_\beta}{2}(M_\beta + 1).$$

The TMFS filter length and polynomial order are estimated as

$$L_\beta = 2\left\lceil \frac{-20\log_{10}\sqrt{\delta_p\delta_s} - 8.4}{30.4(f_s - f_p)} \right\rceil,$$

$$M_\beta = \left\lceil \left(\frac{-20\log_{10}\sqrt{\delta_p\delta_s} - 8.4}{15.2(f_s - f_p)}\right)^{0.7} \right\rceil - 1.$$

The total FIR computation rate is

$$C_{\mathrm{FIR}} = \sum_{k=1}^{K} C_k, \quad \text{where} \quad C_k = \frac{L_k}{2}\frac{N F_{\mathrm{out}}}{\widehat{N}_{k+1}} \approx \frac{D_\infty\left(\frac{\delta_p}{K+1}, \delta_s\right)}{\Delta f_k} F_{\mathrm{in}} \frac{N}{R}\frac{1}{\widehat{N}_{k+1}}.$$

Key observations

- For moderately small R: the optimal structure $K = 1,\ N = 2$
- For larger R: the optimal structure usually has two FIR stages

Optimal factorization algorithm

ret $\leftarrow \emptyset$, result $\leftarrow \infty$
if $R \in \mathbb{N}$ **then**
  AllFactorizations $\leftarrow$ FINDFACTORIZATIONS(R)
  for all $n \in$ AllFactorizations **do**
    $K \leftarrow |n|$, tmp $\leftarrow 0$, product $\leftarrow 1$
    for $k = 1$ to K **do**
      product $\leftarrow$ product $\times\, n[k]$
      tmp $\leftarrow$ tmp + $\frac{D_\infty(\delta_p/K, \delta_s)}{(2 - \text{product}/R \times (1+f_p))/n[k] \times \text{product}}$
    if tmp < result **then**
      result $\leftarrow$ tmp, ret $\leftarrow n$
for $N = 2$ to $\lfloor R \rfloor$ **do**
  AllFactorizations $\leftarrow$ FINDFACTORIZATIONS(N)
  for all $n \in$ AllFactorizations **do**
    $\beta \leftarrow R/N$, $K \leftarrow |n|$
    $l \leftarrow L_\beta(\delta_p/(K+1), \delta_s, 1 - (1+f_p)/(2 \times N))$
    $m \leftarrow M_\beta(\delta_p/(K+1), \delta_s, 1 - (1+f_p)/(2 \times N))$
    tmp $\leftarrow m + l/2 \times (m+1)/\beta$, product $\leftarrow 1$
    for $k = 1$ to K **do**
      product $\leftarrow$ product $\times\, n[k]$
      tmp $\leftarrow$ tmp + $\frac{D_\infty(\delta_p/(K+1), \delta_s)}{(2 - \text{product}/N \times (1+f_p))/n[k] \times \beta \times \text{product}}$
    if tmp < result **then**
      result $\leftarrow$ tmp, ret $\leftarrow n$
return ($F_{\mathrm{in}} \times$ result, ret)

Performance comparison

The table below compares FIR-only systems with the proposed TMFS and FIR approach for $\delta_p = 0.01$, $\delta_s = 0.001$, and $f_p = 0.9$.

R	FIR filters only		TMFS and FIR filters		
	N_k	C_T	β	N_k	C_T
3	3	25.41	3/2	2	29.40
5	5	25.41	5/2	2	18.84
8	4, 2	9.53	4	2	12.90
25	5, 5	7.22	25/2	2	6.17
77	11, 7	4.11	77/4	2, 2	3.36
94	47, 2	3.22	47/2	2, 2	3.11
97	97	25.41	97/4	2, 2	3.08

The gain is most pronounced for prime and poorly factorizable composite ratios. For example, when $R = 97$, the proposed solution reduces the computation rate dramatically.

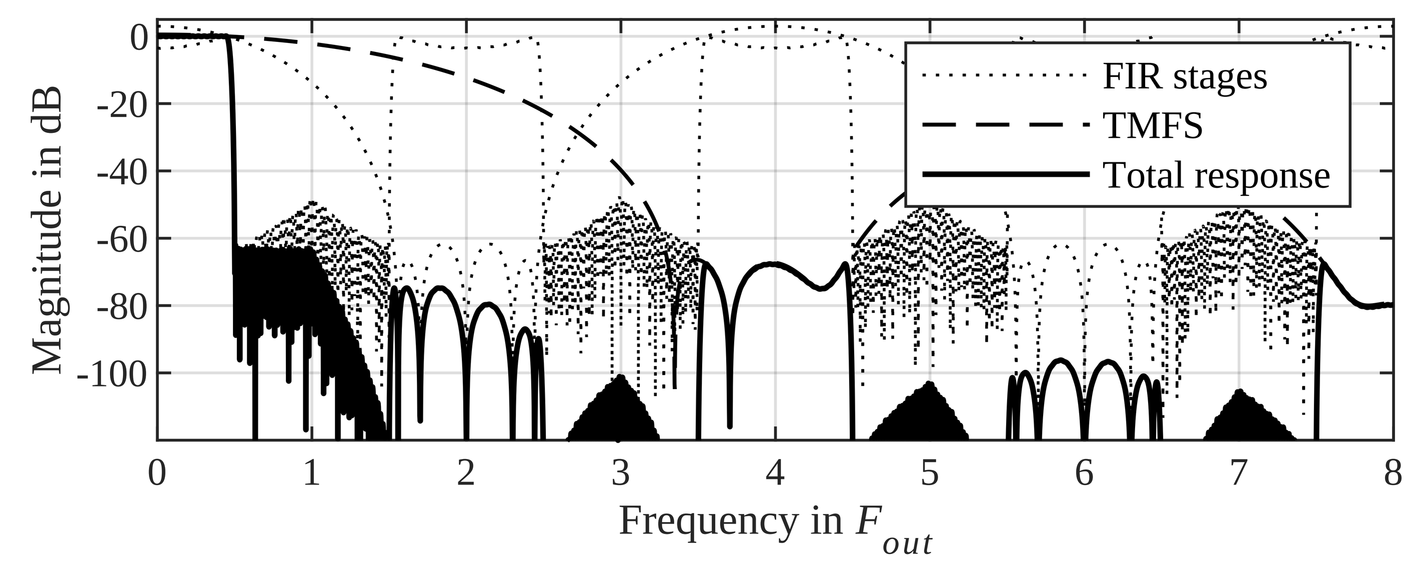

Figure 2. Frequency response of the decimator: FIR stages (dotted), TMFS (dashed), and total response (solid), for the case $R = 97$ from the Table.

Conclusion: Why a cascade of TMFS and FIR filters?

Compared to FIR-only systems:

- TMFS enables arbitrary factorization of R,
- FIR stages work at progressively lower sampling rates,
- the last FIR stage performs passband shaping under milder requirements,
- the overall complexity can be substantially reduced.

The proposed cascade enables substantially more efficient multistage decimation, especially for non-integer, prime, and poorly factorizable ratios.

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